Reyadath Ullah

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Education

2020 - 2025 **B.Sc. in Computer Science and Engineering** **CGPA: 3.73/4.00 (Top 10%)**
University of Dhaka, Bangladesh

Research Interests

- **Federated Learning**, **Reinforcement Learning**, **Edge Intelligence**, **Mobile Computing**, **On Device AI**, **IoT**

Publications

- **[Preprint]** *Data-Diversity-Aware Quantization for Semi-Synchronous Federated Deep Reinforcement Learning in Vehicular Edge Computing*. Ittehad Saleh Chowdhury, **Reyadath Ullah**, Md. Abdur Razzaque.
- *Federated Deep Reinforcement Learning for Optimal Resource Allocation in Vehicular Edge Computing*. **Reyadath Ullah**, Ittehad Chowdhury, Md. Abdur Razzaque, Palash Roy, Mohammad Mehedi Hassan. 13th SAI Computing Conference 2025, London, United Kingdom. [**Paper PDF**]
- *Reducing Complexity in Exporting RMG Products Using Artificial Intelligence in Bangladesh*. **Reyadath Ullah**, Ahmed Shakibul Islam, Toma Ray, Nusrat Najnen Eva. Supply Chain Insider, Volume 12, Issue 01, pp. 61-85.

Research Experience

Remote Research Collaborator, CiteCast Project, William & Mary Sept 2026 - Present

- Collaborating on an IMLS-funded project to identify and preserve web resources referenced in archived US television news, alongside Old Dominion University and the Internet Archive.
- Developing multi-modal machine learning and deep learning models for web citation detection under the supervision of Dr. Alexander C. Nwala and Dr. Yanfu Zhang.

Researcher, Green Networking Research Group, University of Dhaka Sep 2023 - Present

- Co-designed **DA-FeDRL**, a TD3-based scheduler that jointly adapts device selection and bandwidth allocation, improving accuracy while reducing energy use in simulations.
- Developed **DQS-FeDRL**, a framework integrating **data-diversity-aware quantization** and **Bayesian availability-aware scheduling** to optimize communication efficiency and straggler resilience in Vehicular Edge Computing.

Industry Experience

Engineer I, Samsung R&D Institute Bangladesh, Dhaka Aug 2025 - Present

- Migrated the **Samsung Tizen Emulator** backend from legacy QEMU 2.3 to **QEMU 9.2** and redesigned the UI architecture with **SDL2**, replacing Qt to improve cross-platform performance.
- Developed **Tizen Agent Skills (Open Source Repo)**, an open-source Node.js/TypeScript toolkit enabling AI coding assistants (Claude, Gemini, Cline) to automate Tizen SDK workflows via natural language, including app deployment, GDB/CDP remote debugging, and Playwright testing.
- Developed and published the **Tizen VS Code Extension** on the VS Code Marketplace.
- Researching and drafting patents for distributed, federated-learning-based recommendation systems to drive AI feature recommendations across Samsung devices.

Python Developer (Team Lead), Turing (Remote) Dec 2023 - Dec 2024

- Led a distributed team of **6-8 engineers** improving data pipelines for training AI models.
- Built and maintained **Python** and shell automation for large-scale data preparation and deployment.

Python Instructor, CodexPro (Remote) Jan 2023 - Aug 2023

- Taught **basic to intermediate Python** (English batches) to students with diverse backgrounds.
- Helped design and maintain **course materials and assignments** for Python and Scratch.

Projects

Embedded & Systems Projects

Device Drivers for STM32F4xx Microcontrollers **C / C++**

- Developed register-level I^2C, SPI, and UART drivers from scratch for reliable embedded communication without external HALs.

DUOS (RTOS) for ARM Cortex-M4 **C / ARM Assembly**

- Built a real-time operating system featuring priority-based scheduling, context switching, and synchronization primitives.

Software Projects

CoLab **FastAPI, Python, React**

- Engineered the REST API backend and data models for a collaborative project-sharing platform.

DormBuddy **Python, Spring Boot, React**

- Implemented core listing, search flows, and backend logic for a student hostel-finding application.

Other Notable Work

- **CSEDU App** (Flutter/Firebase), **Project Z** (retro 2D C/C++ games via SDL), and Arduino-based sensor prototypes.

Skills

Programming	C/C++, Python, Java, JavaScript, ARM Assembly, Dart
Systems & Tools	Linux/macOS, Git, STM32/embedded C on ARM Cortex-M4, basic Arduino prototyping, socket programming
Frameworks & Data	PyTorch, Flower (federated learning), React, Flutter, Spring Boot, JavaFX, SDL; PostgreSQL, MongoDB, Firebase, SQLite

Competitive Programming & CTF

- **ICPC Asia Dhaka Regional 2024**: 54th place (**Top 17%** nationally); solved **900+** problems across online judges. Profiles: Codeforces, LightOJ, UVA, StopStalk.
- **CTF**: **DU CTF 2023** - 2nd Runner Up; finalist in **EWU National Robofest CTF 2024** and **BUET CTF 2024**.

Standardized Tests

IELTS Academic	Overall: **8.0** (Listening: 9.0, Reading: 8.0, Writing: 7.0, Speaking: 7.5)
GRE General Test	Verbal: **151** (43%), Quantitative: **162** (57%), Analytical Writing: **4.5** (85%)